

DHCP

Kezdés

két hálókártya

```
GNU nano 7.2 /etc/network/interfaces
# This file describes the network interfaces available
# and how to activate them. For more information, see i

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

# Kulso
allow-hotplug ens33
iface ens33 inet dhcp

# Belso
allow-hotplug ens36
iface ens36 inet static
    address 10.0.0.1
    netmask 255.255.255.0
    dns-nameserver 1.1.1.1
```

systemctl restart networking.service

ISC

apt install isc-dhcp-server -y

```
GNU nano 7.2 /etc/default/isc-dhcp-server
# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc
# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.co

# On what interfaces should the DHCP server (dhcpd) serve
# Separate multiple interfaces with spaces, e.g. "et
INTERFACESv4="ens36"
INTERFACESv6=""
```

a belső interfészen legyen, ahova osszon IP-t

A DHCP pool

```
GNU nano 7.2 /etc/dhcp/dhcpd.conf
default-lease-time 86400;
max-lease-time 172800;

authoritative;

subnet 10.0.0.0 netmask 255.255.255.0 {
    range 10.0.0.50 10.0.0.100;
    option routers 10.0.0.1;
    option domain-name-servers 1.1.1.1, 8.8.8.8;
}
```

```
default-lease-time 86400;
max-lease-time 172800;
```

```
authoritative;
```

```
subnet 10.0.0.0 netmask 255.255.255.0 {
    range 10.0.0.50 10.0.0.100;
    option routers 10.0.0.1;
    option domain-name-servers 1.1.1.1, 8.8.8.8;
}
```

```
systemctl restart isc-dhcp-server.service
```

```
systemctl status isc-dhcp-server.service
```

KEA

`apt install kea-dhcp4-server -y`

`nano /etc/kea/kea-dhcp4.conf`

túl sok minden van, benne és nekünk nem kell

két opció

- Ctrl + K és soronként ki törlöd

vagy

- kilépsz, és beírod ezt
`echo "1" > /etc/kea/kea-dhcp4.conf`
így csak egy „1” – est kell törölnöd

Miután üres minden, bemásoljuk a configot

Megjegyzés, másold át előbb **jegyzettömbbe**, szerkeszd meg, és utána másold be

FONTOS, ha szét esik (nincsenek benne a TAB - ok) nem baj, a gép érti, csak nekünk nem tetszik

A **piros** részeket, lehet kell szerkeszteni

Másolás után:

`systemctl restart kea-dhcp4-server.service`

`systemctl enable kea-dhcp4-server.service`

```
{
  "Dhcp4": {
    "interfaces-config": {
      "interfaces": [ "ens36" ]
    },
    "lease-database": {
      "type": "memfile",
      "lfc-interval": 3600
    },
    "valid-lifetime": 86400,
    "max-valid-lifetime": 172800,
    "renew-timer": 43200,
    "rebind-timer": 75600,
    "authoritative": true,

    "subnet4": [
      {
        "subnet": "10.0.0.0/24",
        "pools": [
          { "pool": "10.0.0.50 - 10.0.0.100" }
        ],
        "option-data": [
          {
            "name": "routers",
            "data": "10.0.0.1"
          },
          {
            "name": "domain-name-servers",
            "data": "1.1.1.1, 8.8.8.8"
          }
        ]
      }
    ],
  },
  "loggers": [
    {
      "name": "kea-dhcp4",
      "output_options": [
        {
          "output": "/var/log/kea-dhcp4.log"
        }
      ],
      "severity": "INFO"
    }
  ]
}
```

```
{
  "Dhcp4": {
    "interfaces-config": {
      "interfaces": [ "ens36" ]
    },
    "lease-database": {
      "type": "memfile",
      "lfc-interval": 3600
    },
    "valid-lifetime": 86400,
    "max-valid-lifetime": 172800,
    "renew-timer": 43200,
    "rebind-timer": 75600,
    "authoritative": true,

    "subnet4": [
      {
        "subnet": "10.0.0.0/24",
        "pools": [
          { "pool": "10.0.0.50 - 10.0.0.100" }
        ],
        "option-data": [
          {
            "name": "routers",
            "data": "10.0.0.1"
          },
          {
            "name": "domain-name-servers",
            "data": "1.1.1.1, 8.8.8.8"
          }
        ]
      }
    ],

    "loggers": [
      {
        "name": "kea-dhcp4",
        "output_options": [
          {
            "output": "/var/log/kea-dhcp4.log"
          }
        ],
        "severity": "INFO"
      }
    ]
  }
}
```

NAT

Mind a kettő DHCP szervernél, így:

Kapott DHCP címet, de nincs internet no para broski

Ezeket csak így beírod (ezzel átmenetileg lesz net)

```
echo "1" > /etc/dhcp/dhcpd.conf
```

```
(            apt install iptables -y            )
```

Itt a külső interfész kell

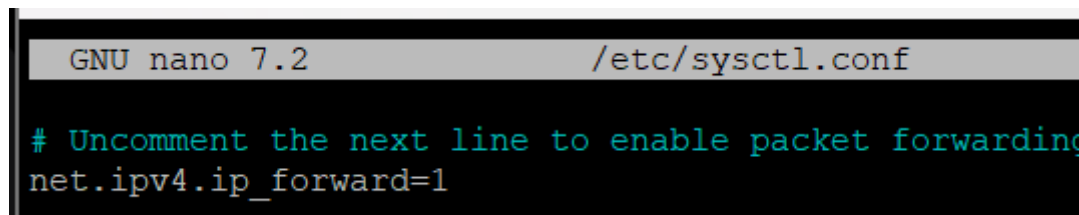
```
iptables -t nat -A POSTROUTING -o ens33 -j MASQUERADE
```

Ha rosszat adtál meg, akkor ez:

```
iptables -t nat -F
```

kitörli, és utána írd be a jót

Ha újraindítás után is szeretnéd, ami egy élő szervernél fontos ugye, akkor:



```
GNU nano 7.2                      /etc/sysctl.conf

# Uncomment the next line to enable packet forwarding
net.ipv4.ip_forward=1
```

Sor elejéről kiveszed a „#” -et

```
sysctl -p                      -> ezzel újraindítod / véglegesíted
```

majd letelepítetted ezt

```
apt install iptables-persistent -y
```

telepítés közben feljön egy ablak, h elmentet-e az eddigi lptables (szabályokat), és YES, YES mindkettő

Eredmény

ISC

```
Command Prompt

Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    Description . . . . . : Intel(R) 82574L Gigabit Network Connection
    Physical Address. . . . . : 00-0C-29-74-8F-A3
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::49d:c6dd:2080:32f9%8(Preferred)
    IPv4 Address. . . . . : 10.0.0.50(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Saturday, May 23, 2026 11:00:20 PM
    Lease Expires . . . . . : Sunday, May 24, 2026 11:00:19 PM
    Default Gateway . . . . . : 10.0.0.1
    DHCP Server . . . . . : 10.0.0.1
    DHCPv6 IAID . . . . . : 83889193
    DHCPv6 Client DUID. . . . . : 00-01-00-01-31-A3-BE-77-00-0C-29-74-8F-A3
    DNS Servers . . . . . : 1.1.1.1
                           8.8.8.8
    NetBIOS over Tcpip. . . . . : Enabled
```

```
Command Prompt

'clear' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Martin>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=27ms TTL=114
Reply from 8.8.8.8: bytes=32 time=26ms TTL=114
Reply from 8.8.8.8: bytes=32 time=24ms TTL=114
Reply from 8.8.8.8: bytes=32 time=23ms TTL=114

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 23ms, Maximum = 27ms, Average = 25ms

C:\Users\Martin>ping google.com

Pinging google.com [192.178.25.174] with 32 bytes of data:
Reply from 192.178.25.174: bytes=32 time=31ms TTL=116
Reply from 192.178.25.174: bytes=32 time=29ms TTL=116
Reply from 192.178.25.174: bytes=32 time=24ms TTL=116
Reply from 192.178.25.174: bytes=32 time=23ms TTL=116

Ping statistics for 192.178.25.174:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 23ms, Maximum = 31ms, Average = 26ms

C:\Users\Martin>
```

KEA

Windows IP Configuration

```
Host Name . . . . . : Gyakorlas
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
```

Ethernet adapter Ethernet0:

```

Connection-specific DNS Suffix . : 
Description . . . . . : Intel(R) 82574L Gigabit Network Connection
Physical Address. . . . . : 00-0C-29-74-8F-A3
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . : Yes
Link-local IPv6 Address . . . . : fe80::49d:c6dd:2080:32f9%8(Preferred)
IPv4 Address. . . . . : 10.0.0.50(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Saturday, May 23, 2026 11:37:04 PM
Lease Expires . . . . . : Sunday, May 24, 2026 11:37:04 PM
Default Gateway . . . . . : 10.0.0.1
DHCP Server . . . . . : 10.0.0.1
DHCPv6 IAID . . . . . : 83889193
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-A3-BE-77-00-0C-29-74-8F-A3
DNS Servers . . . . . : 1.1.1.1
                        8.8.8.8
NetBIOS over Tcpip. . . . . : Enabled

```

```
C:\Users\Martin>
```

```
C:\Users\Martin>ping 8.8.8.8
```

```
Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=29ms TTL=114
Reply from 8.8.8.8: bytes=32 time=28ms TTL=114
Reply from 8.8.8.8: bytes=32 time=25ms TTL=114
Reply from 8.8.8.8: bytes=32 time=23ms TTL=114
```

```
Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 23ms, Maximum = 29ms, Average = 26ms
```

```
C:\Users\Martin>ping google.com
```

```
Pinging google.com [142.251.140.78] with 32 bytes of data:
Reply from 142.251.140.78: bytes=32 time=66ms TTL=116
Reply from 142.251.140.78: bytes=32 time=26ms TTL=116
Reply from 142.251.140.78: bytes=32 time=24ms TTL=116
Reply from 142.251.140.78: bytes=32 time=24ms TTL=116
```

```
Ping statistics for 142.251.140.78:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 24ms, Maximum = 66ms, Average = 35ms
```

C:\Users\Martin>